

CELL DIVISION

Mitosis

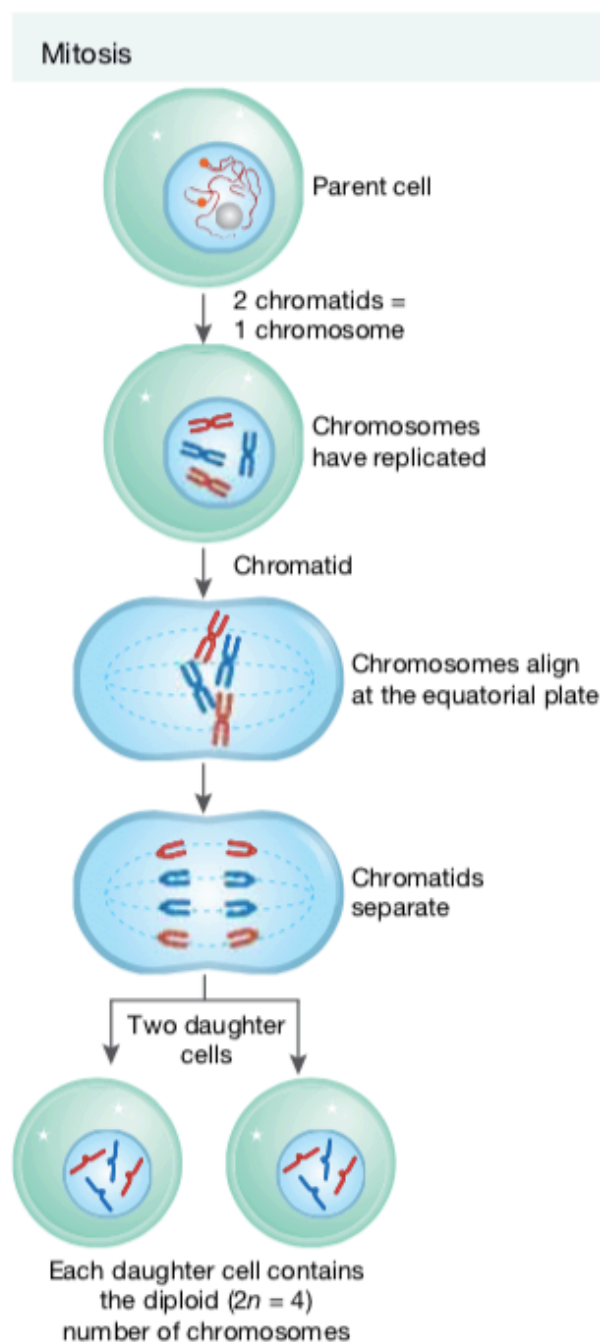
Mitosis produces new cells for **growth** and to **replace damaged cells**.

It occurs for **all cells except for the reproductive cells** (sperm and egg cells).

One cell divides many times and each cell produced will be an **exact copy of the one cell** with **all cells having the same number of chromosomes**.

The nucleus controls the division process.

The steps involved in this process are shown below.



When the chromosomes first become visible and then duplicate themselves, they form **two chromatids** that are joined by a **centromere**.

This still **represents one chromosome only**, not two.



The result of mitosis is that the daughter cells all contain the **normal number of chromosomes**.

Definitions

Diploid Number - normal number of chromosomes.

Haploid Number - half the normal number of chromosomes.

Meiosis

This occurs in the **reproductive organs** and produces **gametes with the haploid number of chromosomes** (half the normal number).

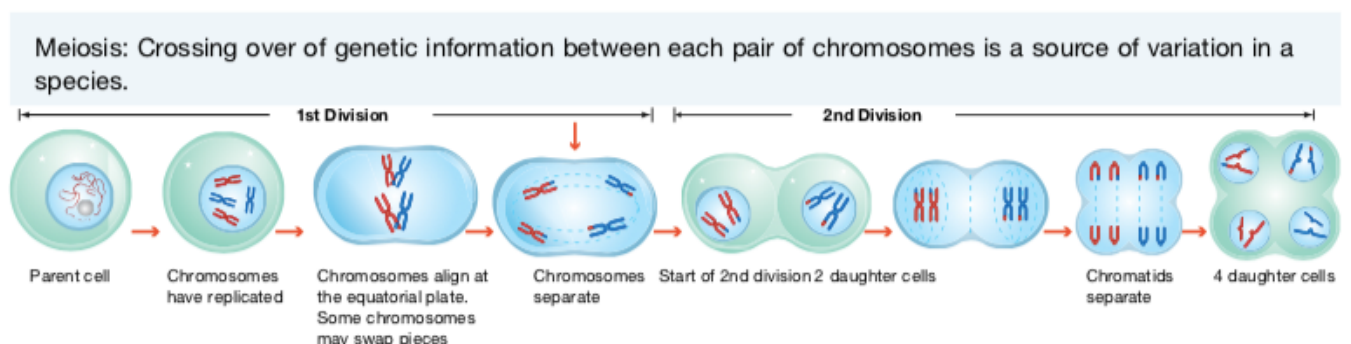
This is important as during the **fertilisation** process between an egg and sperm cell, the **diploid or normal number of chromosomes is restored**.

The process occurs in two stages.

STAGE 1 - the chromosome number is halved.

STAGE 2 - the chromosomes are pulled apart as in mitosis.

The process occurs as shown below.



Definition

Homologous Pair - two chromosomes containing similar genetic information for particular characteristics.